Chapter 4. Linear Regression

Lecture 6-1 (simple and multiple regression, inference, F-test, qualitative predictors, Gauss–Markov, interactions, polynomials)

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1 Simple Linear Regression

Linear regression assumes that the dependence of Y on $X_1, \ldots, X_p$ is linear. True regression functions are never exactly linear, yet the model is extremely useful both conceptually and practically. On the advertising data we may ask: is there a relationship between budget and sales? How strong is it? Which media contribute?

1.1 Model and least squares

$$Y = \beta_0 + \beta_1 X + \varepsilon, \qquad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

β_0 (intercept) and β_1 (slope) are the coefficients. With residuals $e_i = y_i - \hat{y}_i$, the residual sum of squares is $\mathrm{RSS} = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$.

Theorem: least-squares estimates

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Proof. $\partial \mathrm{RSS}/\partial \beta_0 = -2\sum(y_i - \beta_0 - \beta_1 x_i) = 0$ gives $\beta_0 = \bar{y} - \beta_1 \bar{x}$. Substituting, $\partial \mathrm{RSS}/\partial \beta_1 = -2\sum(x_i - \bar{x})\{(y_i - \bar{y}) - \beta_1(x_i - \bar{x})\} = 0$ (using $\sum(x_i - \bar{x}) = 0$) gives $\hat{\beta}_1$. RSS is a convex quadratic, so this is the minimum. ■

On the advertising data, $\widehat{\texttt{sales}} = 7.03 + 0.0475 \times \texttt{TV}$; the line captures the essence of the relationship although it overestimates the trend at the left of the plot.

1.2 Accuracy of the coefficient estimates

With a true line $f(X) = 2 + 3X$, least-squares lines computed from different random samples scatter around it. The standard error measures this variation under repeated sampling:

Theorem: standard errors

If the errors are uncorrelated with common variance $\sigma^2 = \mathrm{Var}(\varepsilon)$ and the x_i are fixed,

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \qquad \mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2 \Big[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\Big].$$

Proof. $\hat{\beta}_1 = \sum_i w_i y_i$ with $w_i = (x_i - \bar{x})/S_{xx}$, $S_{xx} = \sum(x_i - \bar{x})^2$; so $\mathrm{Var}\hat{\beta}_1 = \sigma^2 \sum w_i^2 = \sigma^2/S_{xx}$. Next $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\mathrm{Cov}(\bar{y}, \hat{\beta}_1) = \frac{\sigma^2}{n}\sum w_i = 0$, so $\mathrm{Var}\hat{\beta}_0 = \sigma^2/n + \bar{x}^2\sigma^2/S_{xx}$. ■

σ is estimated by the **residual standard error** $\mathrm{RSE} = \sqrt{\mathrm{RSS}/(n-2)}$. A spread-out design (large S_{xx}) gives a more precise slope.

1.3 Confidence intervals and tests

A $100(1-\alpha)\%$ confidence interval is a range that, under repeated sampling, contains the true parameter with probability $1 - \alpha$:

$$\hat{\beta}_1 \pm t_{1-\alpha/2,\, n-2} \cdot \mathrm{SE}(\hat{\beta}_1) \quad (\approx \hat{\beta}_1 \pm 2\,\mathrm{SE}(\hat{\beta}_1) \text{ for } 95\%).$$

For the advertising data the 95% interval for β_1 is $[0.042, 0.053]$.

Testing H_0: no relationship between X and Y ($\beta_1 = 0$, so $Y = \beta_0 + \varepsilon$) against H_A: $\beta_1 \neq 0$ uses

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)} \sim t_{n-2} \text{ under } H_0 \text{ (Gaussian errors)},$$

and the p-value is the probability of observing $|t|$ or larger.

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

1.4 Overall accuracy: RSE and R^2

$\mathrm{RSE} = \sqrt{\frac{1}{n-2}\sum(y_i - \hat{y}_i)^2}$ estimates the standard deviation of ε and measures lack of fit (in the units of Y).

$$R^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad \mathrm{TSS} = \sum(y_i - \bar{y})^2,$$

the fraction of variance explained, between 0 and 1.

Theorem: $R^2 = r^2$ in simple regression

With $r = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{\sum(y_i - \bar{y})^2}}$, simple linear regression has $R^2 = r^2$.

Proof. $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, and the decomposition $\mathrm{TSS} = \mathrm{RSS} + \sum(\hat{y}_i - \bar{y})^2$ holds (normal equations). So $R^2 = \hat{\beta}_1^2 S_{xx}/S_{yy} = S_{xy}^2/(S_{xx}S_{yy}) = r^2$. ■

For Sales on TV: RSE = 3.26, $R^2 = 0.612$, $F = 312.1$. Actual sales deviate from the true line by about 3,260 units on average; about two thirds of the variability in sales is explained by TV. What counts as a good R^2 depends on the application.

2 Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon, \qquad \texttt{sales} = \beta_0 + \beta_1 \texttt{TV} + \beta_2 \texttt{radio} + \beta_3 \texttt{newspaper} + \varepsilon.$$

β_j is the average effect on Y of a one-unit increase in X_j, *holding all other predictors fixed.*

2.1 Interpreting coefficients

Correlations among predictors cause problems: interpretations become hazardous (when X_j changes, everything else changes) and the variances of all coefficients tend to increase, sometimes dramatically. The ideal is uncorrelated predictors (a balanced design), where each coefficient can be estimated and tested separately. Claims of causality should be avoided for observational data.

Example: the woes of interpreting coefficients (Mosteller and Tukey, 1977)

Y = total money in your pocket, X_1 = total number of coins, X_2 = number of pennies, nickels and dimes. Alone, X_2 is positively associated with Y. But given X_1, more small coins means fewer quarters, so the coefficient of X_2 becomes negative.

2.2 Estimation

The $\hat{\beta}_j$ minimise RSS $= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip})^2$; in matrix form $\hat{\beta} = (X^\top X)^{-1} X^\top y$ (normal equations $X^\top (y - X\hat{\beta}) = 0$), and $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$. With $p = 2$ the fit is the plane minimising squared vertical distances.

	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

For given TV and newspaper budgets, an extra \$1,000 on radio is associated with about 189 additional units sold. The newspaper coefficient is near zero although newspaper alone is significant in a simple regression: newspaper is correlated with radio and acts as a surrogate. Simple and multiple regression coefficients can differ greatly.

3 Four Important Questions

3.1 Is at least one predictor useful?

Test $H_0 : \beta_1 = \cdots = \beta_p = 0$ against H_A: at least one $\beta_j \neq 0$, with

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)} \;\sim\; F_{p,\,n-p-1} \text{ under } H_0.$$

Derivation: why F is the right statistic

Under the linear model, $\mathbb{E}[\mathrm{RSS}/(n-p-1)] = \sigma^2$, and under H_0 also $\mathbb{E}[(\mathrm{TSS}-\mathrm{RSS})/p] = \sigma^2$; if H_0 is false the numerator is larger. With Gaussian errors, $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p-1}$ and $(\mathrm{TSS} - \mathrm{RSS})/\sigma^2 \sim \chi^2_p$ independently under H_0 (Cochran), giving the F distribution. Individual t-tests are not enough: with $p = 100$ and H_0 true, about 5 coefficients will have $p < 0.05$ by chance, while the F-test keeps its level.

Advertising data: RSE $= 1.69$, $R^2 = 0.897$, $F = 570$.

3.2 Deciding on the important variables

Best subsets: fit all subsets and choose by a criterion that balances training error with model size; infeasible for large p (2^p models; over a billion for $p = 40$). Automated searches: **forward selection** starts with no predictors and adds, one at a time, the variable giving the greatest improvement; **backward elimination** starts with the full model and removes the least useful predictor one at a time. Each searches only $1 + p(p+1)/2$ models (details in Ch. 6).

3.3 How well does the model fit?

RSE $= \sqrt{\mathrm{RSS}/(n-p-1)}$, $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}$, and visualisation of residuals. Note R^2 never decreases when predictors are added, while RSE can increase if the decrease of RSS is small relative to the lost degree of freedom.

3.4 How accurate is the prediction?

Three sources of uncertainty: (i) **reducible error** — $\hat{Y} = \hat{\beta}_0 + \cdots + \hat{\beta}_p X_p$ only estimates $f(X) = \beta_0 + \cdots + \beta_p X_p$ (a confidence interval quantifies it); (ii) **model bias** — the linear model is an approximation; (iii) **irreducible error** ε — how much will Y vary from $\hat{Y}$ (a prediction interval, always wider, includes it). For a new x_0: $\mathrm{Var}(x_0^\top \hat{\beta}) = \sigma^2 x_0^\top (X^\top X)^{-1} x_0$ and $\mathrm{Var}(Y_0 - x_0^\top \hat{\beta}) = \sigma^2\{1 + x_0^\top (X^\top X)^{-1} x_0\}$.

4 Qualitative Predictors

Some predictors take a discrete set of values (categorical predictors, factors), e.g. student status or marital status in the Credit data.

4.1 Two levels

For the difference in credit-card balance between house owners and non-owners, define $x_i = 1$ if person i owns a house and 0 otherwise:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{owner,} \\ \beta_0 + \varepsilon_i & \text{non-owner.} \end{cases}$$

Estimates: intercept 509.80 (SE 33.13), own[Yes] 19.73 (SE 46.05, $p = 0.669$). Average debt is \$509.80 for non-owners and \$529.53 for owners; the difference is not significant.

4.2 More than two levels

For region (East, South, West) create two dummies $x_{i1} = \mathbb{I}(\text{South})$, $x_{i2} = \mathbb{I}(\text{West})$:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{South,} \\ \beta_0 + \beta_2 + \varepsilon_i & \text{West,} \\ \beta_0 + \varepsilon_i & \text{East (baseline).} \end{cases}$$

There is always one fewer dummy than levels (with an intercept, K dummies would be collinear with the intercept column). Estimates: baseline East \$531.00; South -18.69 ($p = 0.774$); West -12.50 ($p = 0.826$): no evidence of a regional difference. The coefficients depend on the baseline, but the fitted values and the overall F-test do not.

5 The Gauss–Markov Theorem

Theorem: Gauss–Markov

Let $y = X\beta + \varepsilon$ with $\mathbb{E}\varepsilon = 0$, $\mathrm{Var}\varepsilon = \sigma^2 I$, and X of full column rank. For any fixed c, among all linear unbiased estimators of $\theta = c^\top \beta$, the estimator $\hat{\theta} = c^\top \hat{\beta}$, $\hat{\beta} = (X^\top X)^{-1} X^\top y$, has the minimum variance, and it is unique. Least squares is the **BLUE**.

Proof. *Step 1 (unbiasedness constraint).* A linear estimator is $\tilde{\theta} = a^\top y$. Since $\mathbb{E}(a^\top y) = a^\top X\beta$, unbiasedness for all β requires $a^\top X\beta = c^\top \beta$ for all β, i.e. $X^\top a = c$. *Step 2 (variance comparison).* Least squares corresponds to $a_0 = X(X^\top X)^{-1} c$, which satisfies $X^\top a_0 = c$. Any other unbiased a can be written $a = a_0 + d$ with $X^\top d = 0$. Then $a_0^\top d = c^\top (X^\top X)^{-1} X^\top d = 0$, so

$$\mathrm{Var}(\tilde{\theta}) = \sigma^2 a^\top a = \sigma^2 (a_0^\top a_0 + d^\top d) \;\geq\; \sigma^2 a_0^\top a_0 = \mathrm{Var}(\hat{\theta}),$$

with equality iff $d = 0$. ■

Gauss–Markov needs neither normality nor independence (only uncorrelated, homoscedastic errors), but it restricts to *linear unbiased* estimators: biased estimators such as ridge (Ch. 6) can have smaller mean squared error.

6 Extensions of the Linear Model

The standard model assumes the effects are **additive** and **linear**.

6.1 Interactions

The additive model says that the effect of one more unit of TV is always β_1, regardless of radio. If radio spending makes TV advertising more effective (a *synergy* effect in marketing, an *interaction* in statistics), splitting a \$100,000 budget may beat spending it all on one medium. On the advertising data, residuals show that sales are overestimated when either budget is low and underestimated when it is split.

$$\texttt{sales} = \beta_0 + \beta_1 \texttt{TV} + \beta_2 \texttt{radio} + \beta_3 (\texttt{radio} \times \texttt{TV}) + \varepsilon.$$

	Coefficient	Std. error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

R^2 rises from 89.7% to 96.8%: $(96.8-89.7)/(100-89.7) = 69\%$ of the variability left by the additive model is explained by the interaction. An extra \$1,000 of TV is associated with $(\hat\beta_1 + \hat\beta_3\,\texttt{radio}) \times 1000 = 19 + 1.1\,\texttt{radio}$ units; an extra \$1,000 of radio with $29 + 1.1\,\texttt{TV}$ units.

Hierarchy principle. If an interaction is included, include the main effects too, even if their p-values are not significant: interactions are hard to interpret without main effects, and omitting them changes the meaning of the interaction (and makes the model depend on the origin of the predictors).

6.2 Qualitative–quantitative interactions

For balance on income and student status,

$$\texttt{balance}_i \approx \beta_0 + \beta_1 \texttt{income}_i + \begin{cases} \beta_2 + \beta_3 \texttt{income}_i & \text{student} \\ 0 & \text{non-student} \end{cases}$$

$$= \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3)\texttt{income}_i & \text{student,} \\ \beta_0 + \beta_1 \texttt{income}_i & \text{non-student :} \end{cases}$$

different intercepts *and* slopes for the two groups.

6.3 Nonlinear effects: polynomial regression

On the Auto data, mpg versus horsepower is curved, and $\texttt{mpg} = \beta_0 + \beta_1 \texttt{hp} + \beta_2 \texttt{hp}^2 + \varepsilon$ fits better: estimates 56.9001, -0.4662, 0.0012, all with $p < 0.0001$. It is still a *linear* model (linear in the coefficients), fitted by least squares with $\texttt{hp}^2$ as an extra predictor. Residuals versus fitted values show a strong U-shape for the linear fit (nonlinearity) and little pattern for the quadratic fit.

Summary

- OLS minimises RSS; standard errors, t-tests and confidence intervals quantify the uncertainty of coefficients; RSE and R^2 measure fit.
- The F-test checks whether any predictor is useful; subset searches decide which.
- Dummy variables encode qualitative predictors relative to a baseline.
- Gauss–Markov: OLS is the best linear unbiased estimator.
- Interactions (with the hierarchy principle) and polynomial terms relax additivity and linearity while staying within least squares.